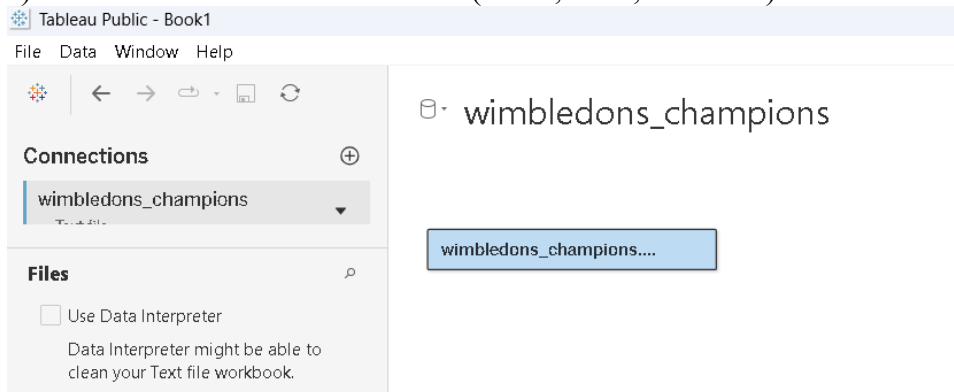


Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

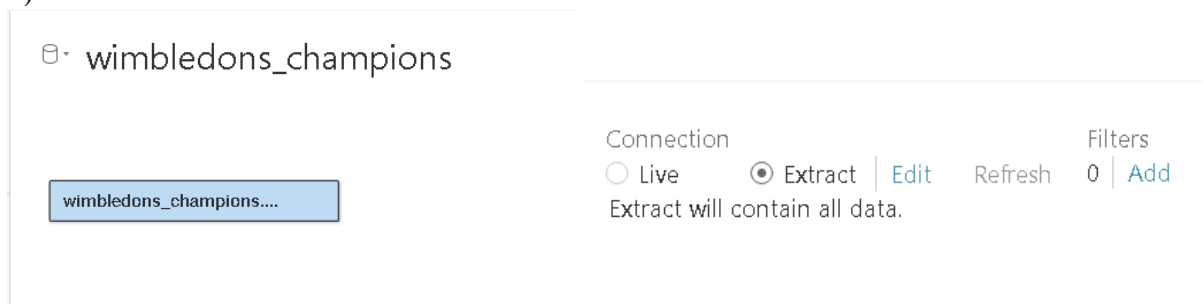
Practical – II

Roll No.: T018	Name: Sachin Ramesh Mehta
Class: TYIT	Batch: 1
Date of Assignment: 31/07/2026	Date/Time of Submission: 31/07/2026

a) Connect to different data sources (Excel, CSV, Database)



b) Create and use data extracts vs live connections



Connection		Filters	
☒ Live	☐ Extract	0	Add
Extract will contain all data.			

wimbledons_champions.csv		12 fields 253 rows		100 rows	
Name	wimbledons_champions.csv	Gender	Champion	Mins	Runner-up Nationality
Description	No description available.				
Fields	Type	Field Name	Phys...	Rem...	
Abc	Gender	wimbl...	Gender		
Abc	Champion	wimbl...	Cham...		
#	Mins	wimbl...	Mins		

Men's	G.L. Patterson	null	AUS	AUS
Men's	G.L. Patterson	null	GBR	AUS
Men's	N.E. Brookes	null	GBR	AUS
Men's	N.E. Brookes	null	NZL	AUS
Men's	J.R. Borotra	80	FRA	FRA
Men's	J.R. Lacoste	null	FRA	FRA

wimbledon_champions.csv 12 fields 253 rows 100 rows

Name wimbledon_champions.csv

Description No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Gender	wimbl...	Gender
Abc	Champion	wimbl...	Cham...
#	Mins	wimbl...	Mins

Abc	Abc	#	Abc	Abc
wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv
Gender	Champion	Mins	Runner-up Nationality	Champion Nationality
Men's	G.L. Patterson	null	AUS	AUS
Men's	G.L. Patterson	null	GBR	AUS
Men's	N.E. Brookes	null	GBR	AUS
Men's	N.E. Brookes	null	NZL	AUS
Men's	J.R. Borotra	80	FRA	FRA
Men's	J.R. Lacoste	null	FRA	FRA

c) Manage metadata (rename, hide, change data types)

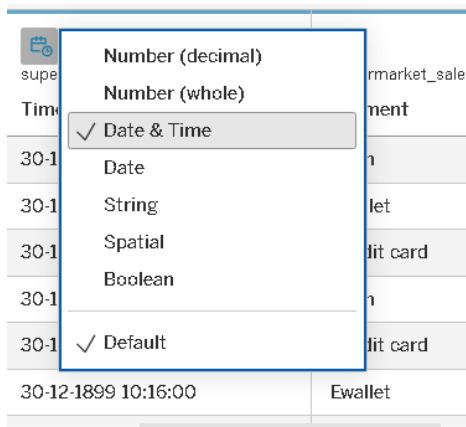
rename :

The screenshot shows the Tableau interface with the 'Tables' pane on the left. The 'Tax 5%' field is selected, and a context menu is open with the 'Rename' option highlighted. The 'Marks' card on the right shows 'Automatic' as the default mark type.

hide:

The screenshot shows the Tableau interface with the 'Tables' pane on the left. The 'Invoice ID' field is selected, and a context menu is open with the 'Hide' option highlighted. The 'Marks' card on the right shows 'Automatic' as the default mark type.

Change data types:



d) Perform joins (inner, left, right) on multiple tables

Inner Join:

wimbledon_champions.csv is made of 2 tables. ⓘ

wimbledon_champions.csv supermarket_sales - supermarke...

Join

Inner ☒ Left ☐ Right ☐ Full Outer ☐

Data Source		supermarket_sales...
Gender	=	Gender (supermarket sal...

Add new join clause

wimbledon_champions.csv 29 fields 0 rows

Name wimbledon_champions.csv

Description No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Gender	wimbl...	Gender
Abc	Champion	wimbl...	Cham...
#	Mins	wimbl...	Mins

Abc	Abc	#	Abc	Abc
wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv
Gender	Champion	Mins	Runner-up Nationality	Champion Nationality

Left Join:

wimbledon_champions.csv is made of 2 tables. ⓘ

wimbledon_champions.csv supermarket_sales - supermarke...

Join

Inner ☐ Left ☒ Right ☐ Full Outer ☐

Data Source		supermarket_sales...
Gender	=	Gender (supermarket sal...

Add new join clause

wimbledon_champions.csv 29 fields 253 rows

Name wimbledon_champions.csv

Description No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Gender	wimbl...	Gender
Abc	Champion	wimbl...	Cham...
#	Mins	wimbl...	Mins

Abc	Abc	#	Abc	Abc
wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv
Gender	Champion	Mins	Runner-up Nationality	Champion Nationality
Men's	G.L. Patterson	null	AUS	AUS
Men's	G.L. Patterson	null	GBR	AUS
Men's	N.E. Brookes	null	GBR	AUS
Men's	N.E. Brookes	null	NZL	AUS
Men's	J.R. Borotra	80	FRA	FRA
Men's	J.R. Lacoste	null	FRA	FRA

Right Join:

wimbledon_champions.csv is made of 2 tables. ⓘ

wimbledon_champions.csv

Join

Inner Left Right Full Outer

Data Source

Gender

supermarket_sales - supermarke...

Gender (supermarket sal...)

Add new join clause

wimbledon_champions.csv 29 fields 1000 rows 100 rows

Name wimbledon_champions.csv

Description No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Gender	wimbl...	Gender
Abc	Champion	wimbl...	Cham...
#	Mins	wimbl...	Mins

Abc	Abc	#	Abc	Abc
wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv	wimbledon_champions.csv
Gender	Champion	Mins	Runner-up Nationality	Champion Nationality
null	null	null	null	null
null	null	null	null	null
null	null	null	null	null
null	null	null	null	null
null	null	null	null	null
null	null	null	null	null

e) Apply data blending between two sources

Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Data Analytics

supermarket_sales - super...

wimbledon_champions

Search

Tables

- Branch
- City
- Customer type
- Date
- Gender
- Invoice ID
- Payment
- Product line
- Time

Pages

Columns Gender

Rows Gender (supermarket ..

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Sheet 1

Gender	Gender
Gender (..	Null
Female	Abc
Male	Abc

f) Implement different types of filters (dimension, measure, date)

Filter [Branch] [X]

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

☒ A
☐ B
☐ C

All None ☐ Exclude

Summary
Field: [Branch]
Selection: Selected 1 of 3 values
Wildcard: All
Condition: None
Limit: None

Reset OK Cancel Apply

Filter Field [Gross Income] [X]

How do you want to filter on [Gross Income]?

- # All values
- # Sum
- # Average
- # Median
- # Count
- # Count (Distinct)
- # Minimum
- # Maximum
- # Standard deviation
- # Standard deviation (Population)
- # Variance
- # Variance (Population)
- # Attribute

Next > Cancel

Filter [Gross Income] [X]

Range of values At least At most Special

At least

5,057.1605 5,057.1605

5,057.1605 5,057.1605

Show: Only Relevant Values ☐ Include Null Values

Reset OK Cancel Apply

Filter Field [Date] [X]

How do you want to filter on [Date]?

- Relative Date
- Range of Dates
- Years
- Quarters
- Months
- Days
- Week numbers
- Weekdays
- Month / Year
- Month / Day / Year
- Individual Dates
- Count
- Count (Distinct)
- Minimum
- Maximum
- Attribute

Next > Cancel

g) Optimize performance using extracts and filtering

Extract Data

Expand All

✓ Data Storage

☒ Logical Tables

Store data using one table for each logical table.

Use this option if you need to use extract filters, aggregation, or other extract settings.

☐ Physical Tables

Store data using one table for each physical table.

> Filters

No filters applied [Edit](#)

> Aggregation

No data aggregated [Edit](#)

> Number of Rows

All rows [Edit](#)

> Incremental Refresh

Not enabled [Edit](#)

Need help with extract settings? [Learn more](#)

History...

Hide All Unused Fields

Cancel

Save Settings

